CoRe-SCD: A Boundary-Task Co-Refinement Network for Semantic Change Detection in Remote Sensing Images

Zhengfang Lou, Xiaoping Lu, Yao Lu, Siyi Li, Guosheng Cai, Kailun Wang, and Bing Liu

***Abstract*—Semantic change detection (SCD) in high-resolution remote sensing imagery is a core technology for quantifying land-cover transitions and supporting refined territorial management. Most existing SCD methods explicitly decouple the task into semantic segmentation (SS) and change detection (CD). However, this paradigm overlooks the inherent synergy between the two tasks in feature representation and optimization objectives, leading to ambiguous boundary localization and weak task interactions. To address these issues, we propose the boundary-task collaborative refinement network (CoRe-SCD). First, a boundary refinement module (BRM) is designed to explicitly inject edge priors into the CD branch, enabling precise dual refinement of spatial structures and semantic contours in changed regions. Second, a cross-task interaction (CTI) module is proposed to model spatiotemporal dependencies via a cross-attention mechanism, achieving bidirectional feature enhancement and strict decoupling of semantic representations from change cues. Finally, a unified multitask loss function is constructed to prevent over-optimization and maintain a strict equilibrium among the subtasks. Comprehensive experiments on two public datasets confirm that collaborative refinement SCD (CoRe-SCD) significantly outperforms other state-of-the-art (SOTA) methods, achieving an optimal balance between predictive accuracy and computational efficiency.**

***Index Terms*—Convolutional neural network (CNN), remote sensing, semantic change detection (SCD), semantic segmentation (SS).**

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I. Introduction

LAND-USE/LAND-COVER change (LULCC) is a critical indicator for characterizing terrestrial ecological processes, resource utilization intensity, and the frequency of human disturbances. Accurate extraction of spatial locations, change magnitudes, and category conversions is essential for decision-making in downstream tasks, including cultivated land protection [1], urban expansion control [2], ecological restoration assessment [3], and territorial spatial planning enforcement [4]. Change detection (CD), which converts radiometric differences into surface condition changes by quantitatively comparing bitemporal remote sensing images, has become a key technology linking Earth observation data with practical application needs. The accuracy of CD directly governs the reliability and effectiveness of subsequent policy implementation.

In the early stages of CD research, due to limitations in model representation capabilities, Binary CD (BCD) was predominantly adopted. While BCD effectively delineates changed regions, its limited representational capacity results in the absence of semantic attributes in bi-temporal images. To compensate for this, early Semantic CD (SCD) studies typically constructed change vectors based on spectral, textural, and color features to first identify regions of change. Subsequently, semantic categories before and after changes were determined on a pixel-by-pixel basis using independent semantic classifiers [5], [6], [7]. Compared to BCD, SCD provides a much more comprehensive representation of change information. As illustrated in Fig. 1, SCD more effectively supports application scenarios requiring explicit identification of "from-to" change trajectories.

In recent years, deep learning has been extensively applied to SCD, significantly enhancing model representational capabilities. End-to-end architectures dominate modern CD tasks, effectively mitigating missed detections and false alarms associated with the cascaded structures of traditional BCD, thereby substantially reducing systematic errors [8]. Deep learning-based SCD architectures are primarily categorized into three types: single-branch, dual-branch, and multitask structures [9]. Among these, multitask structures enable simultaneous processing of semantic and change information from bi-temporal imagery, facilitating precise localization of change regions and efficient classification of their semantic categories [10].

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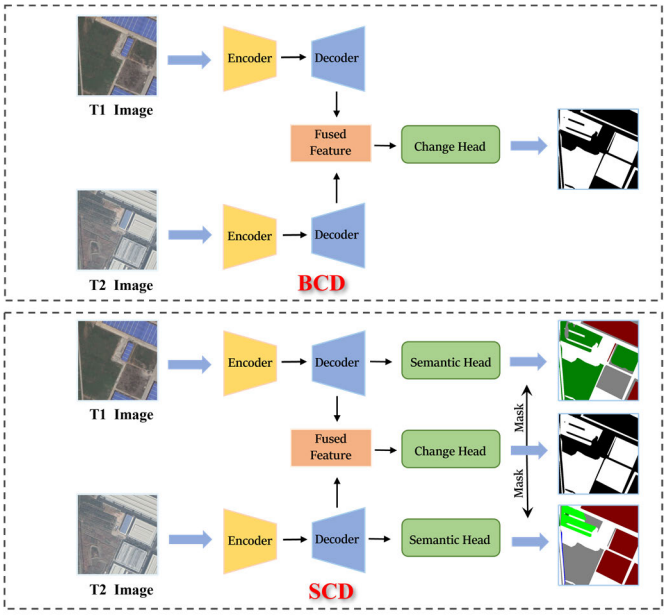


Fig. 1. Architecture diagrams of BCD and SCD.

These architectures typically integrate task-specific feature extraction modules to extract semantic and change features, respectively. Additionally, they can flexibly implement weight-sharing mechanisms across feature extraction branches based on task requirements. Consequently, multitask frameworks provide critical technical foundations for further enhancing the precision of SCD.

Despite this progress, multitask learning-based SCD still faces two major challenges in practical applications.

1) *Insufficient Geometric Accuracy Due to the Absence of Boundary Modeling:* Existing methods often lack explicit modeling of the boundary structures of changed regions, resulting in blurred boundaries between semantic features of different categories. Consequently, this compromises the geometric accuracy of change region localization, inducing false positives commonly referred to as internal false alarms.
2) *Task-Level Constraints Due to Inadequate Feature Interaction:* Current networks typically couple the semantic segmentation (SS) and CD subtasks through simple parallel structures, lacking sufficient cross-task feature interaction. This naive design overlooks the inherent contradiction between intraclass irrelevant differences and category-discriminative features, leading to mutual interference during feature learning. As a result, the performance of both subtasks often becomes mutually constrained, hindering true collaborative optimization.

To address the absence of explicit boundary modeling and insufficient feature interaction in existing multitask SCD methods, this article proposes a joint modeling approach driven by multitask collaborative refinement. We establish efficient collaborative refinement mechanisms and cross-task feature interaction pathways that significantly enhance the accuracy and robustness of SCD, particularly in localizing change region boundaries and detecting internal regions. The primary contributions are as follows.

1) We propose a novel collaborative refinement SCD (CoRe-SCD) network. By constructing a closed-loop framework integrating geometry, semantics, and temporal differences within a unified feature space, CoRe-SCD elevates SCD from traditional parallel task execution to a cross-task mutual interpretation paradigm. This enables precise delineation of change region boundaries, effective suppression of false positives in internal regions, and consistent optimization of semantic labels.
2) To address the issue of insufficient boundary modeling, we introduce a geometry-gated boundary refinement module (BRM). This module leverages boundary probability maps to guide channel-wise feature refinement. It significantly improves the detection accuracy of change boundaries and effectively suppresses "internal false alarms" without increasing inference costs.
3) To overcome the weak feature interaction between subtasks, we design a cross-task interaction (CTI) module. It enhances cross-task complementarity through a bidirectional alignment mechanism for semantic and change information, further improving the model's detection robustness in complex scenarios.

II. Related Work

This section reviews related work in the field of CD, focusing on the evolution from BCD to SCD and the transition from simple feature fusion to complex task coupling strategies.

A. From Binary to SCD

BCD played a pivotal role in the early development of remote sensing image CD [11], [12]. Early BCD methods—including algebra-based [13], [14], transformation-based [15], classification-based [16], advanced models [17], [18], and artificial neural networks [19]—provided diverse solutions. However, with the rapid increase in remote sensing data volume and growing application demands, their limitations have become evident. The primary objective of BCD methods is solely to distinguish between changed and unchanged areas, ignoring the semantic relationships between different land-cover types. Consequently, these methods remain highly susceptible to noise interference and perform poorly in complex surface change scenarios [20].

To address the absence of semantic information in BCD, SCD emerged to provide "from-to" type information, fulfilling the requirements of downstream applications for change category identification. Early SCD research primarily adopted two technical approaches.

1) *Postclassification Comparison:* This approach separately performs SS on bitemporal images, then compares label differences pixel by pixel. While its advantage lies in not requiring paired change samples, systematic errors from the independent classification of the two phases often manifest as real semantic changes, increasing false annotations and reducing detection reliability.

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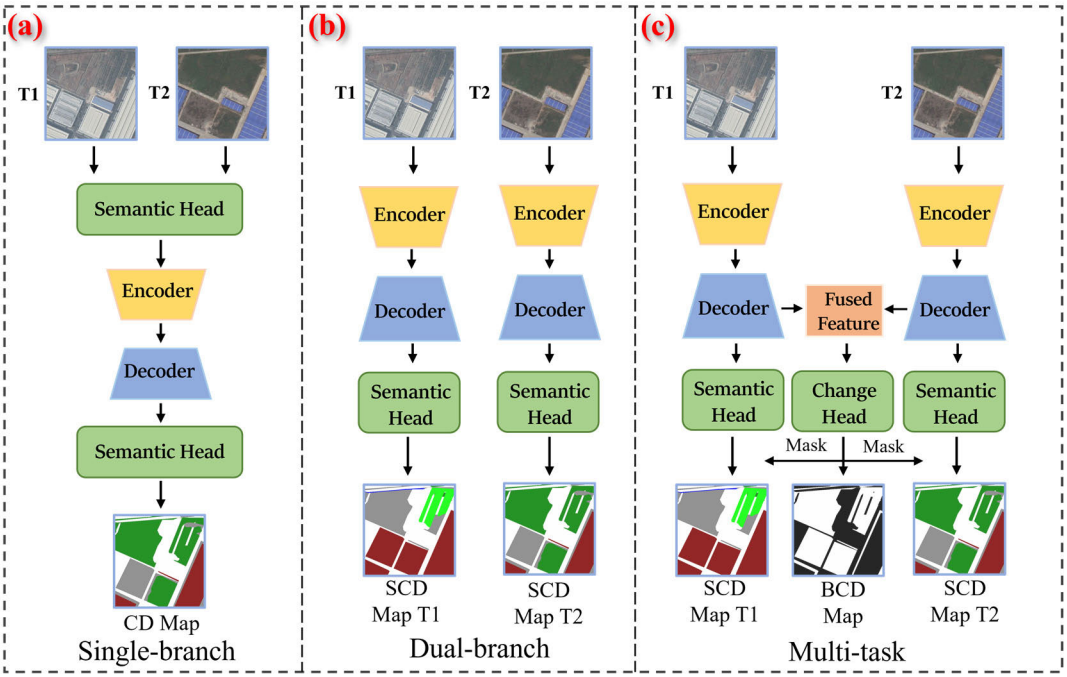


Fig. 2. Overview of three mainstream SCD architectures. (a) Single-branch. (b) Dual-branch. (c) Multitask.

2) *Bitemporal Image Fusion Classification:* This method directly fuses bi-temporal images and performs semantic classification on the fused data. However, it requires training samples proportional to the square of the number of land-use/land-cover (LULC) categories; limited samples cause class confusion and imbalance, severely restricting accuracy. Additionally, it relies on handcrafted features, failing to handle nonlinear distributions in complex scenes. Recently, convolutional neural networks (CNNs) have significantly improved the accuracy of SCD through strong feature extraction capabilities. However, as monitoring tasks increase in complexity, the research focus has shifted from simple feature representation to the structural design of multitask coordination.

B. Multitask Architectures and Task Coupling Paradigms

Deep learning enables end-to-end model training and effectively mitigates cascade errors [8]. Based on architectural differences, mainstream SCD models fall into three primary categories: single-branch, dual-branch, and multitask architectures (see Fig. 2).

The core concept of the single-branch structure is to first fuse bi-temporal images and then perform detection using a single network. Researchers typically synthesize difference images through concatenation or pixel-wise differencing and subsequently input them into pretrained networks, such as DeepLab V3 [21], U-Net [22], and ResNet [23], for change feature extraction. This approach reuses pretrained weights, thereby reducing training complexity; however, fixed fusion methods are prone to losing critical semantic features or introducing noise, which compromises detection accuracy.

To retain independent temporal features, dual-branch architectures process bi-temporal images separately before fusion. Branch weights in these structures can be configured flexibly in either shared or independent modes. Shared weights enhance cross-temporal recognition capabilities and reduce the number of parameters. For example, the DSIFN network proposed in [24] employs a weight-sharing feature extraction scheme and achieves feature fusion via a serial connection approach. However, fusion methods in dual-branch architectures require careful design; otherwise, they may fail to highlight change-sensitive features effectively.

Multitask architectures have emerged as a prominent research hotspot and the mainstream state-of-the-art (SOTA) paradigm for recent SCD [25], [26]. These architectures simultaneously optimize SS and CD tasks through a shared shallow backbone network, leveraging task symbiosis to mutually enhance performance and improve target boundary delineation. The cascade process of segmentation first, then CD is compressed into a single forward inference by this paradigm, and semantic and regional changes are done within a unified framework, thus enhancing the overall accuracy (OA) and efficiency [27], [28], [29]. However, existing multitask SCD methods generally fall into the following three limited paradigms: 1) parallel coupling, where branches operate independently; 2) one-way guidance, where semantic features enhance CD without feedback; and 3) auxiliary boundary constraints, where boundary detection (BD) acts only as a passive loss term. To overcome these limitations, we observe that both SS and CD are highly sensitive to high-frequency details in target boundaries; explicit introduction of a BD task establishes a geometric bridge between semantics and change, significantly suppressing false positives induced by edge blurring [30], [31]. Based on this observation, we propose a BRM to construct

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TABLE I
COMPARISON OF TASK-COUPLING STRATEGIES IN DIFFERENT SCD MODELS

Paradigm	Representative Models	Interaction Mechanism	Coupling Depth
Parallel	HRSCD , BiSRNet	Independent branches with late-stage fusion	Shallow
Auxiliary	BT-SCD	Boundary as auxiliary loss constraint	Medium
One-way Guidance	SCanNet, EGMS-Net	Semantic features enhance change branch	Medium
Co-Refinement	CoRe-SCD (Ours)	Physical gating and bidirectional feedback	Deep

closed-loop interactions among SS, CD, and BD. Edge priors from BD actively constrain high-frequency responses, while SS category confidence inversely suppresses false positives in CD.

C. Global Context Modeling and Local Geometric Constraints

To address the receptive field limitations of standard CNNs, attention mechanisms have been widely adopted in recent SCD architectures. For instance, DSIA U-Net [32] enhances spatial details through deep-shallow feature interactions, while AER U-Net [33] employs multiscale residual attention to improve segmentation robustness. Furthermore, state space models (SSMs) like ChangeMamba [34] and RS-Mamba [35] have demonstrated significant potential in long-range dependency modeling. Although these models excel at capturing global contexts, the explicit preservation of local geometric consistency across tasks—specifically, ensuring that semantic boundaries and change contours are logically aligned—remains a challenge.

In contrast to methods focusing solely on global enhancement, CoRe-SCD introduces a closed-loop co-refinement mechanism to establish bidirectional feedback between subtasks (see Table I). Our approach effectively resolves logical alignment issues while maintaining high computational efficiency.

III. PROPOSED CORE-SCD

The design of the CoRe-SCD is described in this section. The overall network topology is first presented to elucidate the multitask information flow and feature reuse strategy. The core modules designed for task collaboration are then analyzed sequentially. Finally, a multisupervised coupled loss function is introduced to facilitate end-to-end joint optimization.

A. Model Overview

While mainstream SCD methods typically decouple the task into SS and CD, this work explicitly introduces a BD task to construct a unified SS-CD-BD three-task collaborative architecture. The three tasks share a single encoder but use independent decoders, significantly boosting multiscale change recognition accuracy and boundary delineation.

The complete architecture of the CoRe-SCD is illustrated in Fig. 3. To enhance structural transparency, we employ a distinct color-coding scheme: the shared Siamese backbone (cream) facilitates joint feature learning, while the green, orange, and blue branches represent the BD, CD, and SS tasks, respectively. This visualization ensures a clear distinction between shared representations and task-specific refinements. Notably, while the BD task primarily provides explicit geometric constraints for the CD branch via the BRM, it also facilitates feature learning for the SS task through implicit gradient synergy. Since the boundary head is attached to the initial stages of the shared encoder, its gradients force the backbone to prioritize geometry-aware low-level representations during backpropagation. This shared knowledge enhances the edge-delineation capability of the SS task, ensuring that semantic boundaries align logically with detected change contours. For bi-temporal image processing, the model adopts a ResNet-34 backbone pretrained on ImageNet [36]. To preserve high-resolution details, we set the stride of the last two residual modules to 1 and apply dilated convolutions (with dilation rates of 2 and 4), resulting in an output stride of 8. Specifically, given an image $I \in R^{H\times W\times 3}$, the encoder generates a feature pyramid $\{f_1, f_2, f_3, f_4\}$ with spatial resolutions of $\{H/4, H/8, H/8, H/8\}$ and channel dimensions of $\{64\,128\,256\,512\}$. Subsequently, the top-layer feature f_4 is reduced via a 1×1 convolution to $X \in R^{128\times H/8\times W/8}$, which serves as the shared representation for both the SS and CD branches. As previously discussed, a strict weight-sharing Siamese encoder is utilized for bi-temporal images. This design guarantees symmetric representations while avoiding parameter duplication, maintaining an efficient model size of 24.43 M. In Fig. 3, $\{f_1, f_2, f_3, f_4\}$ represent independent feature tensors from this shared encoder, rather than separate weights. Additionally, to accurately model temporal-specific distributions, the semantic decoders utilize independent parameters.

The BD task operates as an auxiliary branch attached to the encoder, utilizing multilevel differential features and raw backbone features to capture both structural textures and change cues. Specifically, the inputs to the boundary head include the bitemporal concatenated features $f_{cat} = [f_1^{t1}, f_1^{t2}]$ and the differential features $\Delta f_i = |f_i^{t1} - f_i^{t2}|$. First, a 1×1 convolution is used to unify the number of channels of Δf_i to 64, and all Δf_i are upsampled to the spatial resolution of f_1 (i.e., $H/4$). Subsequently, the features from all levels are concatenated, and through processes such as convolutional fusion and upsampling, a boundary logit $M_b \in R^{1\times H\times W}$ is generated. Finally, supervision is performed using a class-balanced binary cross-entropy (BCE) loss with boundary labels. This design ensures that the encoder receives direct supervision from the boundary task, allowing the backpropagated gradients to refine low-level geometric features that benefit all three tasks. The

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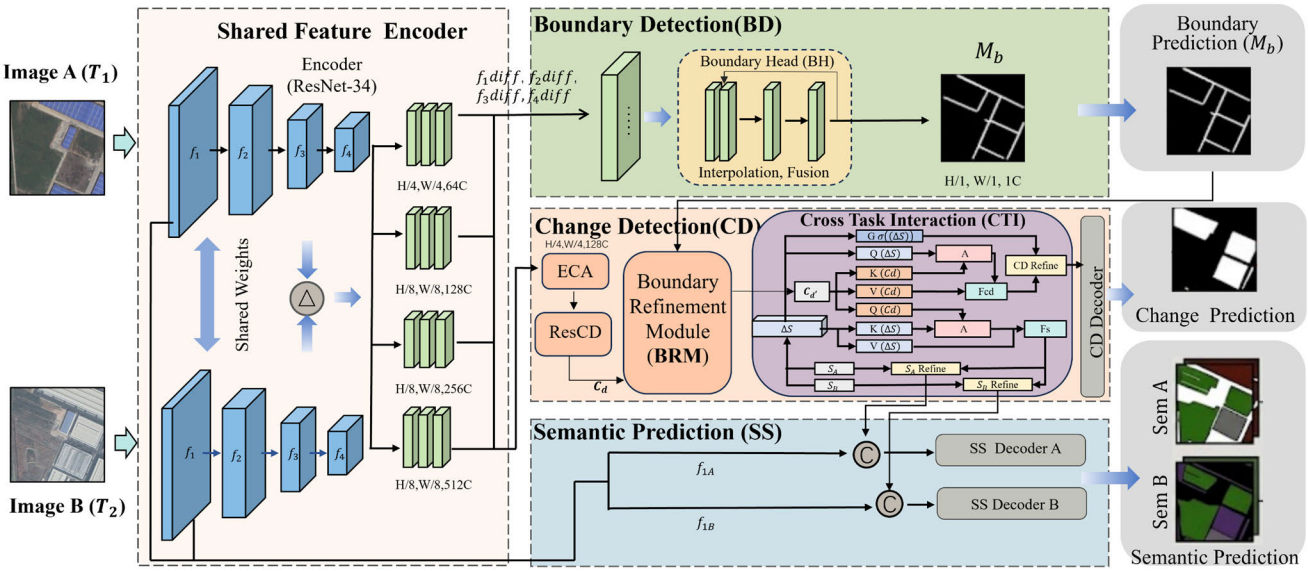


Fig. 3. Architecture of the CoRe-SCD. The framework is color-coded by functional branches: cream for the shared Siamese encoder, green for the BD task, orange/purple for the CD task and task interaction, and blue for SS. Regarding the operational symbols, "©" denotes feature concatenation, and "Δ" represents pixel-wise difference.

resulting boundary probability $\sigma(M_b)$ is then input to the BRM to gate the change features.

In the CD task, the dimension-reduced features f_4' of the two time phases and their absolute difference are first concatenated along the channel dimension, resulting in a tensor with a resolution of $H/8$. Through a channel attention mechanism and the ResCD module—which consists of six stacked standard residual blocks designed to encode the 384-channel concatenated inputs into a robust 128-channel representation—change-sensitive features are extracted, and $Cd \in R^{128\times H/8\times W/8}$ is output. Next, the BRM applies spatial gating weighting to Cd using the boundary probability $\sigma(M_b)$ to highlight the boundaries of change regions. The weighted features, after enhancement by the CTI module, are jointly fed into the decoder along with the skip connection $concat(f_1^{t1}, f_1^{t2})$. Finally, the CD result is output with a size of $1 \times H \times W$.

In the SS task, for each time phase, the semantic branch feeds the dimension-reduced feature $f_4' \in \mathrm{R}^{128\times H/8\times W/8}$ into the CTI module for enhancement, obtaining refined high-level features. Subsequently, the decoder utilizes these features, and the skip connection $f_1 \in R^{64\times H/4\times W/4}$ is used as decoding inputs. The decoder first upsamples the high-level features to *H/4* via transposed convolution, concatenates them with $f1$, fuses them through a 3×3 convolution, and then upsamples the result to the original resolution. After processing by the semantic classifier, the output SS results for each time phase are of size $N \times H \times W$, where N is the number of land cover categories.

B. Boundary Refinement Module

Most existing SCD methods focus primarily on spectral differences and lack explicit modeling of geometric edges. This deficiency leads to false positives under interferences such as seasonal variations, illumination fluctuations, and shadow shifts, along with rough localization of change boundaries. To address this issue, we explicitly introduce BD as an auxiliary task within a unified framework. Using boundary confidence as a geometric prior, we propose the BRM to refine the boundaries of change regions. Specifically, the BRM leverages boundary probability maps to generate edge priors, utilizes channel attention mechanisms to discriminate features, and applies spatial gating to suppress nonboundary disturbances. This achieves explicit enhancement of change edges and robust suppression of false positives. Fig. 4 illustrates the detailed architecture of the BRM.

The core implementation of the BRM proceeds in two steps.

1) *Boundary Prior Acquisition:* Building upon the multiscale feature processing logic of the BD branch, the bi-temporal differential features $\Delta f_1, \Delta f_2, \Delta f_3, \Delta f_4$—representing boundary-sensitive features at different scales—are input into the BD head. To overcome the inevitable feature vanishing of subpixel targets (e.g., narrow roads) in deeply downsampled layers, this process adopts an asymmetric complementary strategy. Specifically, the deep features $(\Delta f_3, \Delta f_4)$ provide the semantic context of changes, while the shallow features $(\Delta f_1, \Delta f_2)$ preserve the high-frequency geometric details necessary for subpixel boundary reconstruction. First, all multiscale features Δf_i are upsampled to the same spatial resolution, then concatenated along the channel dimension, and finally passed through a convolutional layer and a Sigmoid activation function to generate the boundary prediction map $\sigma(M_b)$

$$\sigma(M_b) = Sigmoid\left(Conv\left[Up(\Delta f_1) \,\|\, Up(\Delta f_2) \,\|\, Up(\Delta f_3) \,\|\, Up(\Delta f_4)\right]\right) \quad (1)$$

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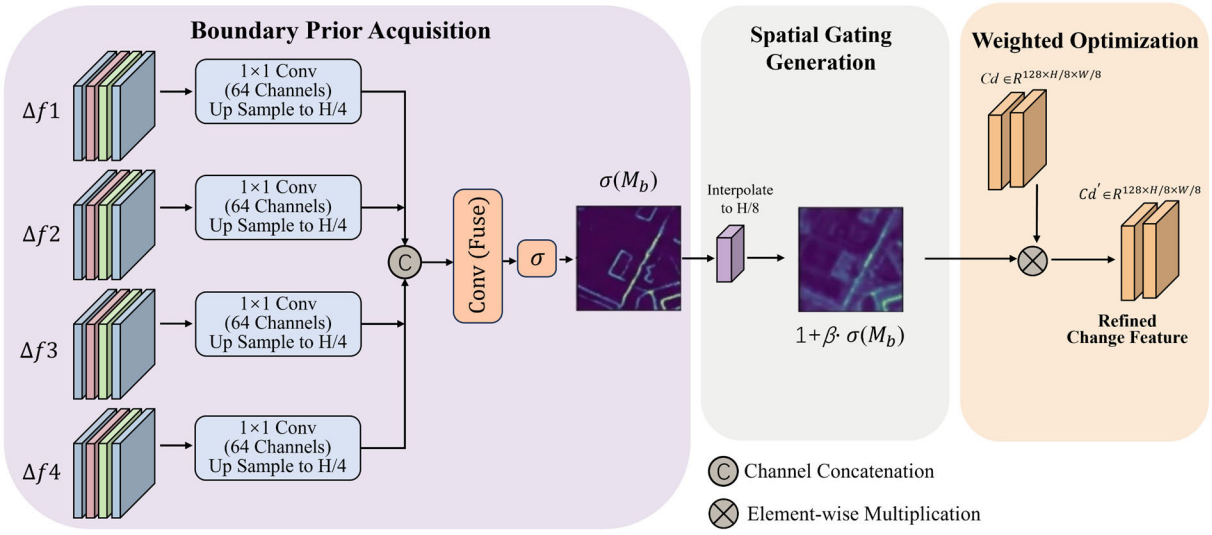


Fig. 4. Architecture diagram of the BRM.

where σ(Mb) (σ denotes the Sigmoid function) is the boundary confidence map required for subsequent weighting. A value closer to 1 indicates a higher probability that the location is a real boundary. Driven by explicit boundary supervision, the network is forced to maintain high sensitivity to minute edge variations in its early layers.

2) *Weighted Optimization of Change Features:* The change features are weighted using the boundary prediction map M_b. Features in boundary regions are assigned higher weights to enhance the model's focus on boundary regions. Assuming Cd' is the output of the CD branch, the feature after boundary weighting can be expressed as

$$Cd' = Cd \odot (1 + \beta \cdot \sigma(M_b)) \tag{2}$$

where β is a learnable parameter. When $\sigma(M_b)$ in boundary regions is close to 1, the change features are significantly amplified, guiding the model to focus on learning boundary details; when $\sigma(M_b)$ is close to 0, the change features remain relatively stable, effectively suppressing pseudo-change interference. Through this spatial gating interaction, the subpixel details from shallow layers successfully compensate for the spatial resolution lost in deeper layers, ensuring the robust and continuous reconstruction of slender targets without being compromised by repeated downsampling.

C. Cross-Task Interaction

SS and CD tasks inherently differ in feature needs: SS focuses on classification accuracy, especially distinguishing foreground/background and categories; CD emphasizes capturing subtle bi-temporal differences and change-region details. Independent feature learning causes SS to ignore change details and CD to lack semantic constraints, leading to pseudo-change misjudgment or class confusion. To solve this, we design the CTI module to enable SS-CD feature interaction: using SS semantic knowledge to constrain CD, reducing nonsemantic interference like illumination fluctuations and CD change information to enhance SS, and improving change-region classification. This achieves bidirectional "semantics-change" and "change-semantics" injection in a unified feature space. To ensure the efficiency of feature interaction, all projection layers in CTI adopt a bottleneck design with a reduction ratio of $r = 8$, minimizing computational overhead while capturing essential cross-task correlations. As shown in Fig. 5, the core of the CTI module comprises three parallel paths: difference gating, semantic-difference cross-attention, and change-semantic feedback.

1) *Difference Gating Path:* It mainly suppresses noise in nonchange regions by constructing a difference gating map G. This path leverages prominent differences in the semantic space to detect potential change regions. The semantic difference map $\Delta S = |S_A - S_B|$ is fed into a 1×1 convolutional layer, and the resulting difference gating map G is used to initially suppress background noise in nonchange regions. Denoting the dimension-reduced features f_4' as the semantic feature maps S_A, S_B of the former and latter time phases, respectively, the semantic difference map ΔS and the difference gating map G can be expressed as

$$G = \sigma\left[conv_{1\times1}(\Delta S)\right], \quad \Delta S = |S_A - S_B|. \tag{3}$$

2) *Semantic–Difference Cross-Attention Path:* In this path, the difference map ΔS is treated as the query, $Q = Conv_Q(\Delta S)$, while the change features Cd' serve as the key and value, $K = Conv_K(Cd')$ and $V = Conv_V(Cd')$. To reduce computational cost while preserving spatial resolution, all projection layers are implemented using 1×1 convolutions and employ a bottleneck design with a reduction ratio of $r = 8$ (i.e., the channel dimension is reduced from 128 to 16 and then restored). The module generates attention weights via a serial spatial–channel

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hybrid mechanism: first, channel information is aggregated through global average pooling and max pooling, and a multilayer perceptron (MLP) produces the channel attention weights $A_c \in R^{C\times1\times1}$; subsequently, based on the channel-refined features, a convolution layer with kernel size 7×7 generates the spatial attention weights $A_s \in R^{1\times H\times W}$. Finally, the combined attention map A_c is applied to the feature V through pixel-wise element-wise multiplication $\odot$ via broadcasting. The refined change features F_{cd} are defined as

$$\begin{cases} A_c = \sigma\left(MLP\left(AvgPool\left(Q\right) + MaxPool\left(Q\right)\right)\right) \\ A_s = \sigma\left(Conv_{7\times7}\left(\left[AvgPool\left(A_c \odot V\right); \right.\right.\right. \\ \qquad\qquad \left.\left.\left. MaxPool\left(A_c \odot V\right)\right]\right)\right) \\ F_{cd} = G \odot \left(A_s \odot A_c \odot V\right). \end{cases} \tag{4}$$

3) *Change-Semantic Feedback Path:* In this path, to accurately feed the structural information of the change regions back to the semantic branch, the change features Cd are treated as the query, $Q = Conv_Q(Cd)$, while the semantic difference map ΔS serves as the key and value, $K = Conv_K(\Delta S)$ and $V = Conv_V(\Delta S)$. As illustrated by the reinforced pathways in the purple region of Fig. 3, the change attention submodule takes the change-sensitive feature C_d as the primary query source. By establishing this explicit connection from the CD branch back to the SS interaction space, the model can effectively propagate temporal variance cues to refine categorical boundaries. Consistent with the semantic–difference path, the projection layers for Q, K, and V are implemented using 1×1 convolutions and incorporate a bottleneck design with a reduction ratio of $r = 8$ to optimize parameter efficiency. This path reuses the serial attention mechanism: first, channel attention is computed to capture cross-task category correlations, followed by spatial attention to precisely localize change boundaries. The resulting combined attention weights are applied to the value V via broadcasting to generate the semantic feedback map $F_s \in R^{C\times H\times W}$. Subsequently, F_s is used through residual connections to enhance the bi-temporal semantic features S_A and S_B. To improve training stability and control feedback strength, a *tanh* activation is applied for nonlinear mapping, and a learnable scaling factor α (initialized to 0.1) is introduced. The refined semantic features S'_A and S'_B can be expressed as

$$\begin{cases} F_s = A_s \odot A_c \odot \mathrm{V} \\ S'_A = S_A + \alpha \cdot \tanh\left(F_s\right) \\ S'_B = S_B + \alpha \cdot \tanh\left(F_s\right). \end{cases} \tag{5}$$

D. Loss Function

This study uses four loss functions to train CoRe-SCD, including SS loss L_{seg}, CD loss L_{change}, semantic consistency loss L_{sc}, and boundary loss L_{bd}.

1) The SS loss is a multiclass cross-entropy loss between the predicted category probability distributions of the two time phases and the ground-truth category labels. The calculation method of L_{seg} for each pixel is

$$L_{seg} = -\frac{1}{N}\sum_{i=1}^{N} q_i \log\left(p_i\right) \tag{6}$$

where p_i denotes the predicted probability, q_i denotes the ground-truth label, and N denotes the number of categories.

2) The CD loss is a BCE loss between the predicted binary change map and the ground-truth label map. To alleviate the problem of imbalanced proportions between unchanged and changed pixels, this article adopts a weighted BCE loss. The calculation method of L_{change} is

$$L_{change} = -\frac{1}{N}\sum_{i=1}^{N}\left(\omega_1 \cdot y_c \log\left(p_c\right) + \omega_0 \cdot \left(\left(1-y_c\right)\log\left(1-p_c\right)\right)\right) \tag{7}$$

where ω_1 and ω_0 are the weights of positive and negative samples, respectively; p_c and y_c represent the predicted change probability and ground-truth label, respectively.

3) The purpose of the semantic consistency loss L_{sc} is to enhance the consistency between CD regions and SS results. This loss function is calculated based on cosine contrastive learning. It operates solely in the semantic feature space and does not depend on category labels. The calculation method of L_{sc} is

$$L_{sc} = \begin{cases} 1 - \cos\left(S_A, S_B\right), & y_c = 0 \\ \max\left[0, \cos\left(S_A, S_B\right)\right], & y_c = 1 \end{cases} \tag{8}$$

where S_A and S_B are the feature vectors of the SS results for the two time phases, respectively.

4) In the boundary loss L_{bd}, there is an imbalance in the pixel ratio between unchanged and changed boundaries. For its calculation here, a weighted BCE loss is still used. Furthermore, to prevent task imbalance where the BD task might dominate training, we maintain L_{bd} as an auxiliary constraint rather than a primary objective. As verified by our sensitivity analysis in Section V-C, this balanced weighting ensures that the model achieves a synergistic co-refinement without compromising the stability of categorical feature extraction or leading to gradient explosion in the shared encoder.

5) *The Final Total Loss of CoRe-SCD Can be Expressed as*

$$L = \omega_1\left(L_{seg1} + L_{seg2}\right) + \omega_2 L_{change} + \omega_3 L_{sc} + \omega_4 L_{bd} \tag{9}$$

where L_{seg1} and L_{seg2} are the represent the respective bi-temporal SS losses. The average of these two losses is used to represent L_{seg}. The cross-entropy and cosine similarity have been normalized to the same numerical range, and the learnable parameters α and β within the model already account for partial dynamic task balancing. Consequently, the weight coefficients $(\omega_1, \omega_2, \omega_3, \omega_4)$ are set to (0.5, 1.0, 1.0, 1.0), respectively.

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IV. Dataset Description and Experimental Settings

A. Datasets

Experiments selected two public SCD benchmarks, SECOND and Landsat-SCD. Covering different spatial resolutions, geographic scenarios, and change type counts, they fully validate the model's detection performance in complex heterogeneous scenes. The specific specifications and preprocessing methods of the two datasets are as follows.

1) Second Datasets: The SECOND dataset is a large-scale, high-quality benchmark of high-resolution remote sensing imagery in the SCD field [37]. Images sourced from multiplatform sensors cover urban areas in Chinese cities such as Hangzhou, Chengdu, and Shanghai, comprising a total of 4662 bi-temporal image pairs. Each image is uniformly sized at 512 × 512 pixels with a spatial resolution ranging from 0.5 to 3 m, facilitating the capture of fine-scale urban land cover changes. The annotation system encompasses six land cover categories: nonvegetated surfaces, trees, low vegetation, water bodies, buildings, and playgrounds. Based on real-world category transition relationships, 30 distinct change types are derived, with pixel-level semantic and change labels provided for all images. The dataset is divided into training (2968 pairs), validation (847 pairs), and test (847 pairs) sets at a 3:1:1 ratio, with consistent scene distributions and change-type proportions maintained across subsets to mitigate data bias in model evaluation.

2) Landsat-SCD Datasets: The Landsat-SCD dataset [38] collects Landsat images with a spatial resolution of 30 m from the Xinjiang region of China between 1990 and 2020. The original data include 8468 pairs of bi-temporal images. After filtering out the subsets that have undergone spatial enhancement in the original dataset, 2385 pairs of valid images without enhancement interference are finally retained. Each image in this dataset is cropped to a size of 416 × 416 pixels and has a spatial resolution of 30 m, making it suitable for capturing macroscale land cover changes at the regional level (e.g., conversion between farmland and desert, expansion of construction land). The annotations include five categories: no-change class, farmland, desert, buildings, and water bodies, from which a total of ten change types are derived. The overall proportion of changed pixels in the dataset is 18.89%. In the experiment, the dataset is divided into training (1431 pairs), validation (477 pairs), and test (477 pairs) at a ratio of 3:1:1.

B. Evaluation Metrics

To comprehensively and objectively evaluate the performance of the CoRe-SCD model in the SCD task, we conduct evaluations using six metrics: OA, Separated Kappa (SeK), mean intersection over union (mIoU), precision (Pre), recall (Rec), and $F1$-score. The specific definitions and calculation methods of these metrics are as follows.

1) *OA:* As a commonly used metric for segmentation tasks [39], [40], OA measures the proportion of correctly predicted pixels by the model to the total number of pixels. Its calculation relies on the confusion matrix $Q = \{q_{i,j}\}$. The calculation formula is

$$OA = \sum_{i=0}^{N} q_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} \tag{10}$$

where $q_{i,j}$ represents the number of pixels with a ground-truth label of class j and a predicted label of class i (where $i, j \in \{0, 1, \ldots, N\}$), and N is the total number of land cover categories.

2) *mIoU:* To mitigate the interference of class imbalance on evaluation results, in the SCD task, mIoU is obtained by calculating the average of the IoU of change regions (IoU_c) and nonchange regions (IoU_{nc}). This metric can more fairly reflect the model's prediction accuracy across different categories, especially for low-proportion change categories [37]. The calculation formula is

$$\begin{cases} IoU_{nc} = q_{00} \Big/ \left(\sum_{i=0}^{N} q_{i0} + \sum_{j=0}^{N} q_{0j} - q_{00} \right) \\ IoU_c = \sum_{i=1}^{N}\sum_{j=1}^{N} q_{ij} \Big/ \left(\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} - q_{00} \right) \\ mIoU = (IoU_{nc} + IoU_c)/2. \end{cases} \tag{11}$$

3) *SeK:* SeK is an accuracy metric specifically designed for the SS task. Its core feature is eliminating the interference of nonchange pixels on evaluation results, focusing solely on the semantic classification accuracy of change pixels. Mathematically, the exponential term $e^{IoU_c - 1}$ acts as a dynamic modulator. Since the derivative of $e^{IoU_c - 1}$ increases as IoU_c approaches 1, this metric is highly sensitive to performance variations in high IoU regions, effectively amplifying the rewards for high-quality CD. Compared to weighted $F1$-score, SeK provides a more rigorous evaluation of semantic consistency within change regions by decoupling the detection quality from the classification accuracy, thereby avoiding the bias caused by the overwhelming proportion of no-change pixels. This metric enhances the semantic discrimination ability of change regions through IoU_c. The calculation formula is

$$\begin{cases} \rho = \sum_{i=0}^{N} \hat{q}_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij} \\ \eta = \sum_{i=0}^{N} \left(\sum_{j=0}^{N} \hat{q}_{ij} * \sum_{j=0}^{N} \hat{q}_{ji} \right) \Big/ \left(\sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij} \right)^2 \\ SeK = e^{IoU_c - 1} \cdot (\rho - \eta)/(1 - \eta). \end{cases} \tag{12}$$

4) *F1-Score:* $F1$ is a balanced metric that comprehensively considers precision and recall [41]. It is suitable for SCD tasks with class imbalance, as it can simultaneously reflect the model's ability to "reduce pseudo-changes

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(Precision)" and "capture all real changes (Recall)." The calculation formula is

$$\begin{cases} \text{Precision} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=1}^{N}\sum_{j=0}^{N} q_{ij} \\ \text{Recall} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=1}^{N} q_{ij} \\ F_1 = \dfrac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \end{cases} \tag{13}$$

C. Experimental Settings

All experiments were implemented via PyTorch, trained on a single NVIDIA GeForce RTX 4060 GPU. The training hyperparameter configurations are as follows: initial learning rate 0.1 (with decay scheduling), Adam optimizer, batch size 8, and 80 epochs. Following the optimal configuration determined in our sensitivity analysis, the multitask loss was formulated as a weighted sum of the subtask losses (with weights set to 0.5, 1.0, 1.0, and 1.0, respectively). To preserve bi-temporal change consistency, only random horizontal flipping and 90° rotation were used for data augmentation. All comparative models were trained with the above settings to ensure fairness. Finally, the performance of all models was evaluated via OA, SeK, mIoU, and $F1$, and the calculation results of each metric were the average of three independent experiments on the test set to reduce the impact of random errors on the evaluation results.

V. Experimental Results

A. Comparative Experiments

1) Comparative Methods: To comprehensively verify the performance advantages of the proposed CoRe-SCD model, this experiment selects seven mainstream, publicly available SCD methods with accessible code for comparison. A comprehensive evaluation is conducted from both quantitative accuracy and qualitative visualization perspectives. The selected comparative methods include HRSCD-4, BiSRNet, SCanNet, BT-SCD, BT-HRSCD, EGMS-Net, and ChangeMamba, covering different architectural designs and core technologies. A brief introduction to these seven methods is provided below.

HRSCD-4 [42]: Integrates SS and CD into a single multitask network, utilizing difference skip connections to transfer semantic information and strengthen subtask correlations.

BiSRNet [41]: Built on the SSCD-l architecture, it introduces single-temporal and cross-temporal semantic reasoning blocks, alongside a semantic consistency loss, to explicitly model semantic correlations.

SCanNet [26]: Based on a triple encoder–decoder framework, it employs a Semantic Change Transformer (SCanFormer) to explicitly model "from-to" semantic transitions in the token space.

BT-HRSCD [43]: Utilizes an HRNet-based dual-branch encoder to retain high-resolution details, alongside BiFAM and HRDE modules to enrich semantic insights and change precision.

BT-SCD [31]: Introduces BD as an auxiliary task, employing pixel- and logit-level interaction strategies and a bidirectional change feature enhancement (BCFE) module for collaborative optimization.

EGMS-Net [44]: An enhanced multitask Siamese network that employs a change information enhancement module (CIEM) for fine-grained features and a change information guidance module (CIGM) to strengthen the interaction between the CD and SS branches.

ChangeMamba [34]: Introduces the SSM-based Visual Mamba architecture to SCD for efficient global spatial context modeling, featuring tailored decoders with three spatiotemporal modeling mechanisms.

2) Quantitative Results: A quantitative comparative analysis was conducted between the CoRe-SCD model and other comparative models on the SECOND and Landsat-SCD datasets. Due to differences in hardware, the reproduced accuracy of some comparative methods shows slight deviations from the results reported in their original papers. We performed all experiments in the identical GPU environment to ensure strict fairness for horizontal comparisons.

As shown in Table II, the CoRe-SCD model achieved the best performance on both datasets. On the SECOND dataset, our model demonstrates a superior balance in suppressing false positives and capturing true changes in complex urban scenarios. On the Landsat-SCD dataset, the performance advantage of CoRe-SCD is further amplified. Notably, it outperforms the second-best model, EGMS-Net, by 2.76% in SeK and 0.97% in mIoU, while also achieving the highest OA (96.03%). This fully demonstrates the model's generalization capability in macro-regional-scale scenarios.

As presented in Table II, early methods like HRSCD-4 and BiSRNet struggle to effectively integrate semantic and boundary information, yielding suboptimal SeK scores and showing limited cross-scenario generalization. While recent mainstream methods like SCanNet and ChangeMamba improve global contexts through Transformer or state-space architectures, they still fall short in precise edge localization, which restricts their mIoU improvements. Similarly, although the BT series and EGMS-Net attempt to introduce auxiliary boundary tasks or multiscale fusion, their performance continues to lag behind CoRe-SCD due to the lack of deep bidirectional feature interaction between the SS and CD branches.

In contrast, CoRe-SCD incorporates the proposed CTI and BRM modules. This model not only enhances the synergy between the SS and CD tasks via explicit boundary priors but also significantly improves detection performance in complex scenes through bidirectional alignment and task interaction. Ultimately, CoRe-SCD simultaneously ensures the completeness of changed regions and the precision of geometric contours, comprehensively outperforming all comparative models across the board.

3) Qualitative Results: To qualitatively evaluate the model's performance across diverse scenarios, we selected two representative image pairs from the SECOND and Landsat-SCD datasets. The visualization results are presented in Figs. 6 and 7. Overall, the semantic masks generated by

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TABLE II
COMPARISON OF SOTA MODEL ACCURACY ON BENCHMARK DATASETS (%)

ExP.	Model	Landsat-SCD dataset						SECOND dataset					
		mIoU	SeK	F1	Prec	Rec	OA	mIoU	SeK	F1	Prec	Rec	OA
1	HRSCD-4	77.39	30.64	73.16	70.62	75.89	91.08	70.96	18.94	58.81	59.86	57.80	86.39
2	BiSRNet	82.31	42.25	80.90	80.71	81.09	93.68	72.93	22.35	61.91	64.75	59.31	87.73
3	SCanNet	86.50	52.69	85.76	84.85	86.68	95.28	73.06	22.69	62.07	64.22	60.05	87.68
4	BT-HRSCD	86.67	53.07	85.92	85.14	86.73	95.34	72.54	21.42	60.69	62.39	59.08	87.23
5	BT-SCD	86.56	53.36	86.25	83.97	88.67	95.30	72.71	21.89	61.23	63.17	59.40	87.41
6	EGMS-Net	87.34	55.57	87.29	86.56	88.03	95.71	72.73	22.50	62.43	64.49	60.50	87.66
7	ChangeMamba	85.28	49.97	84.70	82.15	87.42	94.78	72.88	22.13	61.68	62.70	60.69	87.38
8	CoRe-SCD	**88.31**	**58.33**	**88.36**	**86.71**	**90.07**	**96.03**	**73.90**	**23.96**	**63.22**	**65.32**	**61.25**	**87.93**

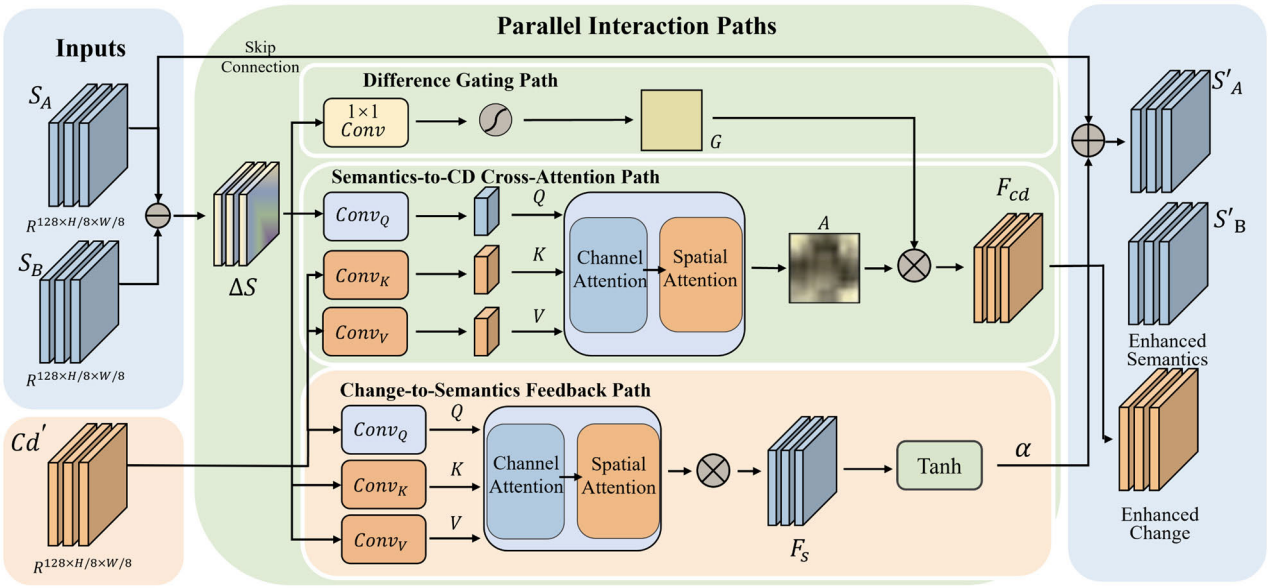


Fig. 5. Architecture diagram of the CTI.

CoRe-SCD consistently outperform those of the comparative methods across three key aspects: boundary sharpness, category consistency, and change region integrity.

In the high-resolution SECOND dataset, complex textures and fragmented land cover pose significant challenges for boundary localization. As demonstrated in the locally magnified views of Fig. 6, baseline methods such as SCanNet often struggle to balance recall and precision. This results in either noticeable false positives along unchanged edges in Fig. 6(a), or missed detections of small-scale objects amid severe color variations in Fig. 6(b). In contrast, by leveraging the cross-task alignment of the CTI module and edge priors from the BRM, CoRe-SCD effectively suppresses background noise and aligns predictions tightly with ground-truth boundaries. This demonstrates its robustness in extracting subtle semantic changes against complex backgrounds.

However, extreme spectral similarity can still induce semantic confusion, as observed between "Ground" and "Low vegetation" in the bottom-left of Fig. 6(b). This limits the BRM's spatial gating and causes localized boundary inaccuracies, highlighting the *a*-direction for future refinement in low-contrast scenarios.

In medium-resolution Landsat-SCD scenes, the mixed-pixel effect further exacerbates edge blurring. As shown in the locally enlarged views of Fig. 7(a), baseline methods lack precision in boundary delineation within the transition zone between farmland and desert, often producing over-smoothed or jagged edges. The results in Fig. 7(b) further corroborate this observation; when processing small-scale building targets, conventional methods frequently exhibit substantial missed and false detections. Through the synergistic operation of its BRM and CTI modules, CoRe-SCD successfully balances CD completeness with segmentation contour accuracy. Even under severe spatial resolution constraints, the model reconstructs clear, closed boundary lines, significantly mitigating aliasing errors in edge regions.

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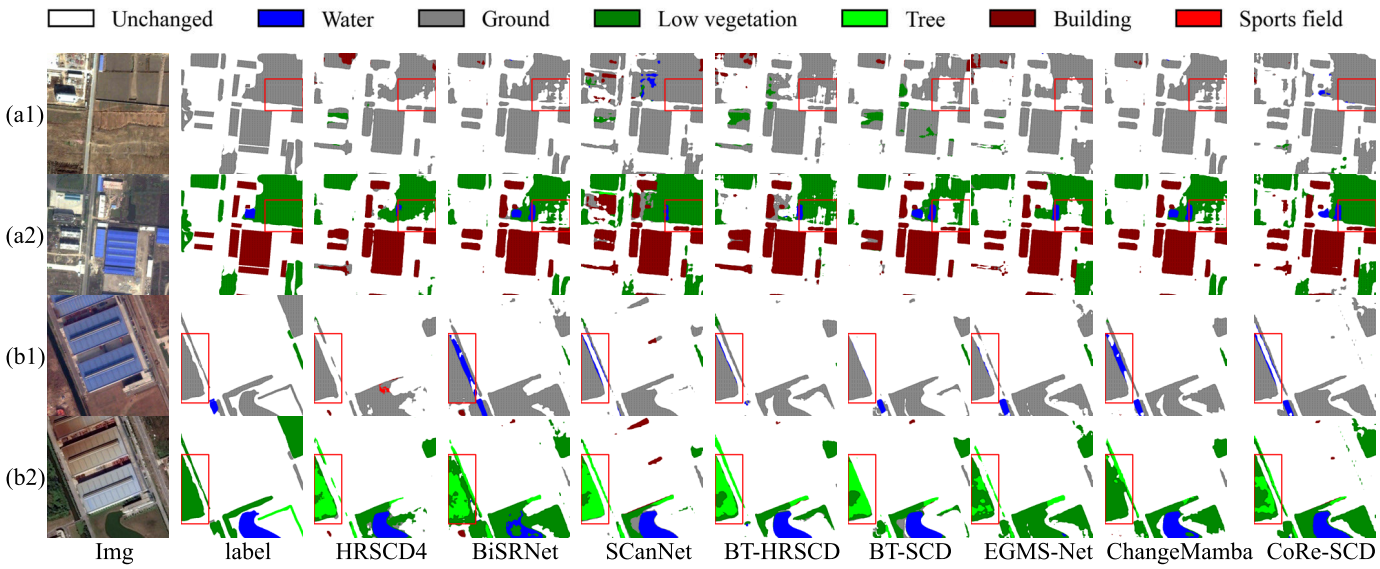


Fig. 6. Visualization analysis on the SECOND dataset. (a1) and (a2) Denote the pre-event and post-event images of the first scene, respectively. (b1) and (b2) Denote the pre-event and post-event images of the second scene, respectively.

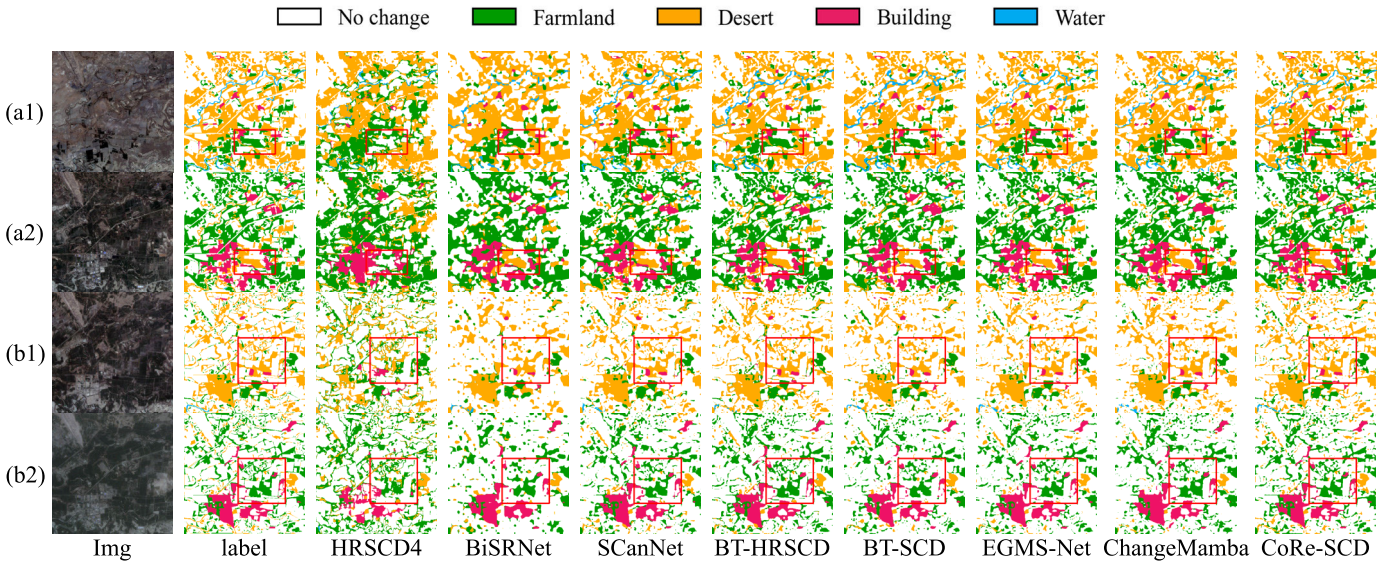


Fig. 7. Visualization analysis on the Landsat-SCD dataset. (a1) and (a2) Denote the pre-event and post-event images of the first scene, respectively. (b1) and (b2) Denote the pre-event and post-event images of the second scene, respectively.

In summary, whether excelling at high-resolution detail restoration or maintaining scale robustness across medium resolutions, CoRe-SCD consistently delivers sharper edges, superior category consistency, and fewer false positives. By harmonizing CD completeness with contour localization accuracy, these results fully validate the broad application potential of the proposed framework in practical remote sensing scenarios.

B. Ablation Experiments

To verify the effectiveness of the proposed components and the logic of intertask collaboration, we conducted a series of ablation experiments on the Landsat-SCD dataset.

1) Effectiveness of Architectural Modules: We first established a baseline multitask SCD network devoid of any interaction mechanisms. Subsequently, we incrementally integrated the CTI and BRM. As shown in Table III (Exp. 9–11, and 8), several key observations emerge:

Adding CTI or BRM alone yields SeK improvements of 2.79% and 0.35%, respectively, confirming their positive guidance for feature refinement. Crucially, integrating both modules simultaneously yields a substantial SeK leap of 6.21% over the baseline. Notably, this synergistic gain is significantly greater than the arithmetic sum of their individual improvements (3.14%). This indicates that the geometric priors from the BRM and the task feedback from the CTI are not merely functionally stacked; instead, they achieve a deep logical coupling. Specifically, the boundary information from the BRM provides the necessary geometric precision for CTI's feedback loop, while the CTI leverages these priors to refine categorical representations.

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TABLE III
ACCURACY EVALUATION TABLE FOR MODEL ABLATION EXPERIMENTS ON BENCHMARK (%)

Exp.	Setting	Backbone	CTI	BRM	FB	GT	SeK	F1	mIoU	OA	Params(M)
9	Baseline	Resnet-34					52.12	85.80	86.13	95.25	24.36
10	w/ BRM only	Resnet-34		√		√	52.47	85.95	86.28	95.35	24.42
11	w/ CTI only	Resnet-34	√				54.91	86.90	87.14	95.55	24.37
12	w/o Feedback (FB)	Resnet-34	√	√		√	53.26	86.22	86.57	95.37	24.43
13	w/o Gating (GT)	Resnet-34	√	√	√		54.53	86.76	87.07	95.54	24.42
8	CoRe-SCD (Full)	Resnet-34	√	√	√	√	58.33	88.36	88.31	96.03	24.43
14	CoRe-SCD(Resnet50)	Resnet-50	√	√	√	√	66.92	91.50	91.14	97.11	27.95

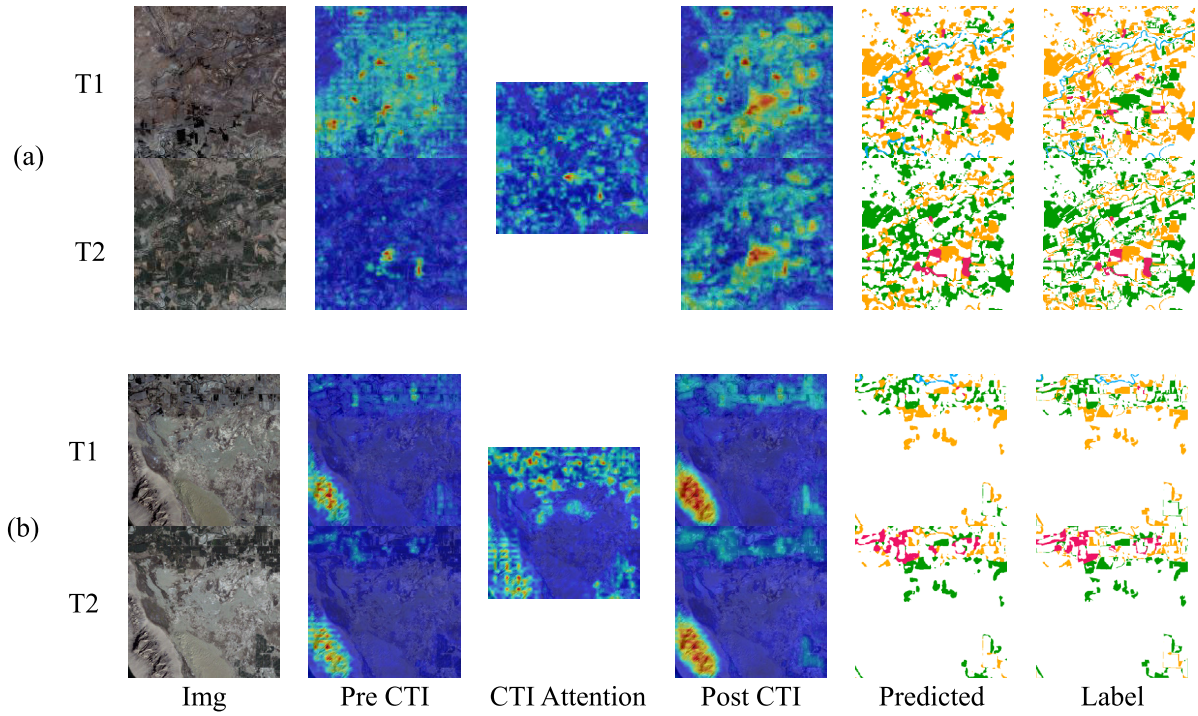


Fig. 8. Visual analysis of the feature evolution process within the CTI module. (a) and (b) Represent two different scenes, respectively.

2) Scalability of the Proposed Architecture: To further investigate the capacity limit and scalability of CoRe-SCD, we upgraded the shared Siamese encoder from ResNet-34 to a deeper ResNet-50 (Exp. 14). As expected, the enhanced feature extraction capability of the deeper backbone yields substantial performance gains, pushing the SeK to 66.92% and the $F1\%$ to 91.50%. This confirms that the proposed collaborative framework is not solely reliant on specific network depths; rather, it is highly scalable and can seamlessly integrate with more powerful backbones to tackle even more complex remote sensing scenarios.

3) Effectiveness of Interaction Paths: To further investigate the internal logic of CoRe-SCD, we conducted path-level ablation experiments targeting the change-to-semantic feedback and geometric gating mechanisms.

Change-to-Semantic Feedback: Disabling the feedback path in the CTI (Exp. 12) results in a significant SeK drop of 5.07%. This validates our hypothesis that back-propagating change information to the semantic branch is essential for resolving boundary ambiguities and maintaining category consistency.

Geometric Gating: Removing the gating mechanism in the BRM (Exp. 13) leads to a 3.80% performance decline. This proves that explicitly leveraging boundary confidence for spatial gating significantly outperforms simple feature fusion in filtering unstructured noise and preserving geometric integrity.

4) Interpretability Analysis of the CTI Module: To investigate how the CTI module suppresses spurious variations, we visualize its feature evolution process using two representative Landsat-SCD scenes (see Fig. 8). In the plain-desert transition zone [see Fig. 8(a)], mixed-pixel interference initially blurs semantic boundaries. However, after applying the CTI attention map, the network accurately identifies true changes, significantly enhancing intraclass homogeneity and preserving sharp contours for narrow targets like rivers.

Similarly, in mountainous areas with severe shadow variations [see Fig. 8(b)], the CTI module successfully breaks the entanglement between actual changes and illumination

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TABLE IV
SENSITIVITY ANALYSIS OF LOSS WEIGHTS ON LANDSAT-SCD (%)

Exp.	Setting	$(\omega_1, \omega_2, \omega_3, \omega_4)$	SeK	F1	mIoU	OA
15	Boundary-enhanced	(0.5, 1.0, 1.0, 2.0)	42.32	80.75	82.43	93.59
16	CD-focused	(0.5, 2.0, 1.0, 1.0)	55.83	87.33	87.48	95.75
17	SS-focused	(1.0, 1.0, 1.0, 1.0)	57.81	88.13	88.15	95.98
8	Balanced (Ours)	(0.5, 1.0, 1.0, 1.0)	58.33	88.36	88.31	96.03

TABLE V
ABLATION STUDY OF SEMANTIC CONSISTENCY LOSS VARIANTS (%)

Exp.	mode	SeK	F1	mIoU	OA
18	MSE	45.94	82.69	83.79	94.14
19	None	43.19	81.41	82.82	93.79
20	Smooth	46.69	82.82	84.30	94.30
8	Paper	58.33	88.36	88.31	96.03

fluctuations. By assigning low attention weights to shadows while maintaining high activations for true changes, it demonstrates robust feature decoupling. The network preserves essential textural features for semantic classification while strictly isolating illumination noise from the CD branch.

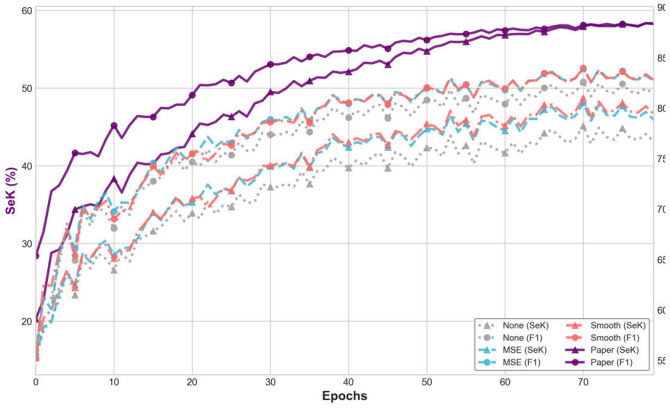


Fig. 9. Convergence curves of different semantic consistency loss variants.

C. Discussion

1) Sensitivity Analysis of Loss Weights: We conducted a sensitivity analysis of loss weight selection on the Landsat-SCD dataset. The default balanced configuration was compared against three task-biased configurations: CD-focused, SS-focused, and boundary-enhanced. The loss weights are denoted as $(\omega_1, \omega_2, \omega_3, \omega_4)$ corresponding to the total loss $L = \omega_1(L_{seg1} + L_{seg2}) + \omega_2 L_{change} + \omega_3 L_{sc} + \omega_4 L_{bd}$.

As illustrated in Table IV, the balanced configuration (0.5, 1.0, 1.0, 1.0) achieves the optimal performance, indicating that maintaining an equilibrium in gradient flow between tasks is crucial for the co-refinement mechanism. Regarding the impact of task bias, explicitly increasing the weights for the CD loss (CD-focused) or the SS loss (SS-focused) leads to a decline in SeK by 2.50% and 0.52%, respectively. This degradation suggests that over-emphasizing a single task disrupts the synergistic interaction among the different branches. Furthermore, doubling the boundary refinement weight (boundary-enhanced) triggers a severe performance drop, with the SeK plummeting to 42.32%. This sharp decline confirms that the boundary task is highly sensitive to high-frequency noise and potential registration errors within the dataset. Therefore, a moderate geometric constraint is indispensable for achieving stable convergence of the model.

2) Analysis of the Semantic Consistency Loss Variants: To validate the design of the semantic consistency loss L_{sc}, particularly the necessity of applying a hard margin operation to regions of change ($y_c = 1$), we conducted detailed ablation experiments on the Landsat-SCD dataset. We compared four variants: 1) None: removes the semantic consistency constraint; 2) mse: employs the standard mean squared error; 3) Smooth: replaces our hard margin with the smooth softplus approximation commonly used in metric learning; and 4) Paper: utilizes the $max(0, cos(S_A, S_B))$ hard margin. Table V summarizes the quantitative results.

TABLE VI
SENSITIVITY ANALYSIS OF HYPERPARAMETERS α AND β IN CTI AND BRM (%)

Exp.	Setting (α, β)	SeK	F1	mIoU	OA
21	(1.0, 0.1)	54.55	86.73	86.59	95.02
22	(1.0, 0.5)	54.12	86.58	86.87	95.49
23	(1.0, 2.0)	53.54	86.31	86.22	94.86
24	(0.1, 1.0)	54.23	86.58	86.82	95.34
25	(0.5, 1.0)	53.82	86.40	86.76	95.40
26	(2.0, 1.0)	53.34	86.24	86.15	94.86
8	(1.0, 1.0)	58.33	88.36	88.31	96.03

Experiments show that introducing semantic consistency constraints significantly improves model performance, but the choice of penalty function is critical. Comparisons reveal that the Smooth variant underperforms the hard-cutoff design used in this article ($max(0, cos(S_A, S_B))$) by 11.64% and 5.54% on the SeK and $F1$ metrics, respectively. Although the quadratic penalty of mse provides a marginal improvement over the "None" configuration, it still significantly lags behind the proposed method. This is because mse imposes excessive gradients on difficult samples, leading to a distorted feature space; meanwhile, the Smooth variant, lacking a zero-gradient cutoff, continuously penalizes the model in an attempt to infinitely increase the feature distances. Both approaches cause the network to expend excessive representational capacity on distance optimization, thereby weakening the effectiveness of boundary extraction and semantic classification. In contrast,

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TABLE VII
COMPARISON OF COMPUTATIONAL COSTS OF SOTA MODELS

Model	Param(M)	FLOPs(G)	FPS(frames/sec)	Latency(ms)
HRSCD-4	13.71	36.73	27.5	36.35
BiSRNet	23.38	380.59	70.1	14.27
SCanNet	27.90	529.88	55.1	18.15
BT-HRSCD	32.97	263.13	10.3	96.71
BT-SCD	24.91	456.5	57.7	17.34
EGMS-Net	23.92	432.16	48.1	20.81
ChangeMamba	54.28	146.80	10.47	95.47
CoRe-SCD(ours)	24.43	410.66	53.9	18.56

the approach selected in this article effectively prevents the over-optimization of the feature space, ensuring an optimized balance among the subtasks. Furthermore, as evidenced by the convergence curves in Fig. 9, the Hard Margin method not only yields superior final accuracy but also exhibits accelerated convergence and greater stability during the initial training phases.

3) Sensitivity Analysis of Module Hyperparameters: To comprehensively evaluate the robustness of the architecture proposed in this article, this section conducts a sensitivity analysis on two core hyperparameters: the scaling factor α in the CTI module and the gating weight β in the BRM. We employed a univariate control strategy, in which one parameter was fixed at its default value of 1.0 while the other was adjusted. The quantitative results on the Landsat-SCD dataset are detailed in Table VI.

The results indicate that model performance peaks when both α and β are set to 1.0, at which point the model achieves its optimal accuracy; deviating from this configuration leads to a significant decline in the SeK metric of approximately 3.8% to 5.0%. Specifically, for the CTI module, a smaller α value results in insufficient CTI, failing to effectively convey semantic difference information to the CD branch. Conversely, an excessively large α value over-amplifies the interaction strength, causing semantic noise to interfere with change discrimination. Similarly, for the BRM, a smaller β value weakens the role of explicit geometric guidance, while an excessively large β value over-amplifies high-frequency edge features, thereby increasing false positives induced by minor registration deviations. In summary, setting $\alpha = 1.0$ and $\beta = 1.0$ ensures the most stable gradient flow and achieves an optimal balance of multitask collaboration.

4) Computational Complexity and Efficiency Analysis: We comprehensively evaluate the practicality of the model by comparing the computational costs of various methods across three dimensions: number of parameters (Param), computational load (FLOPs), and inference speed (FPS and Latency). All evaluations use 512×512 pixel bi-temporal RGB images as input.

As shown in Table VII, CoRe-SCD achieves an optimal balance between performance and efficiency. Compared to heavyweight architectures like ChangeMamba or high-complexity networks like SCanNet, CoRe-SCD significantly reduces the structural burden while maintaining superior change discrimination capability. Meanwhile, when compared to other highly efficient methods such as BT-SCD and EGMS-Net, our model maintains highly competitive inference speeds. These results confirm that CoRe-SCD ensures computational efficiency without compromising accuracy, making it well-suited for application scenarios that require a strict balance between computing resources and inference speed.

VI. CONCLUSION

This article addresses two core challenges in SCD tasks: insufficient boundary localization accuracy and weak feature interactions between subtasks. We propose the CoRe-SCD framework, achieving performance breakthroughs through unified optimization across three stages: encoding, decoding, and interaction. Key innovations are as follows.

1) *Encoding Stage:* We construct a weight-sharing Siamese encoder utilizing a ResNet-34 backbone. By integrating dilated convolutions, the encoder effectively balances the preservation of high-resolution spatial details with the extraction of large-scale contextual changes. Additionally, channel attention mechanisms enhance fine-grained difference features, laying a robust foundation for downstream tasks.
2) *Decoding Stage:* We propose the BRM to explicitly inject geometric edge priors into the CD branch. This design achieves dual refinement of spatial structures and semantic categories, effectively mitigating edge aliasing and suppressing spurious changes within uniform regions.
3) *Interaction Stage:* We introduce the CTI module, which employs cross-attention mechanisms to establish bidirectional "semantic-to-change" and "change-to-semantic" enhancements. Furthermore, by integrating a hard-margin semantic consistency loss,

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the network strengthens the feature discrimination of changed regions, thereby overcoming the limitations of traditional unidirectional task interactions.

Extensive experimental validation on two public datasets (SECOND and Landsat-SCD) demonstrates that CoRe-SCD achieves optimal performance, fully proving its effectiveness and versatility across different resolutions and complex geographic scenarios. To further enhance performance, future work will incorporate a semi-supervised pseudo-label iteration mechanism to augment the model's upper bound using large-scale unlabeled data. Concurrently, teacher–student distillation and structured pruning will be employed to reduce model size, lowering inference latency while maintaining accuracy to meet edge deployment requirements.

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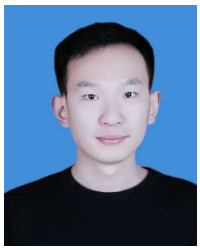
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